EFY TESTED
S.C. Dwivedi

MILLI-OHM METER
With 0.1 To 1-Ohm Range

A. SAMIUDDHIN

Most shunt resistors have values below one ohm. Such low values of resistances cannot be measured reliably due to range and resolution limitation of the measurement devices. These are usually measured using resistance bridges that require a special setup.

Presented here is a circuit that can measure resistances of 0.1-ohm to 1-ohm. EFY lab's prototype of the milli-ohmmeter is shown in Fig. 1.

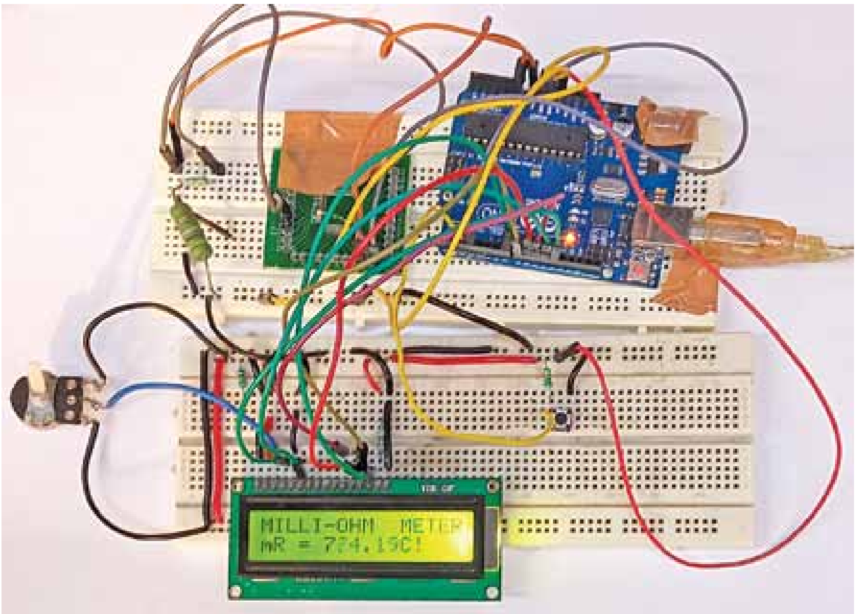

Fig. 1: EFY Lab's prototype on breadboard

Circuit and working

Circuit diagram of the milli-ohm meter is shown in Fig. 2. The circuit is built around Arduino Uno board (Board1), low-dropout regulator MIC5219 (IC1), and 16 × 2 LCD display (LCD1).

When current passes through a resistor, a voltage drop occurs across the resistor in proportion to the current passing through it. By using this principle, known as Ohm's Law, the resistance is measured by using this circuit.

The resistance to be measured is connected across test points A and B in the circuit. The voltage drop across the resistor appears at analogue pin A0 of the Arduino board. The analogue reference voltage is changed to internal 1.1V reference for better resolution.

The Arduino Uno is the brain of this circuit that calculates the resistance based upon the analogue-to-digital (ADC) value and displays it on the LCD. IC1 is configured as a constant-current regulator.

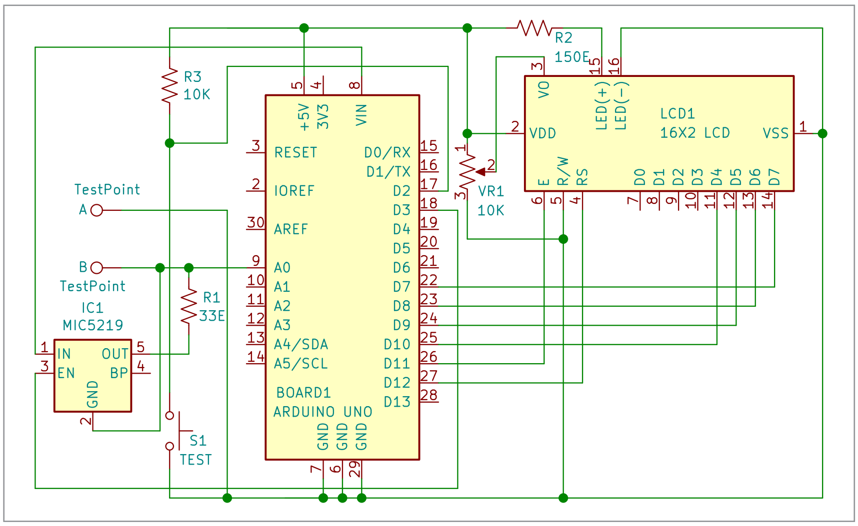

Fig. 2: Circuit diagram of milli-ohm meter

When the push-to-on switch S1 is pressed, IC1 is enabled by the Arduino, and it supplies constant current of 100mA across the test resistor. A voltage drop occurs across the test resistor whose resistance is calculated by the Arduino using Ohm's Law and it is displayed on LCD1.

The resistance to be measured